Technical Supplement: all-orders torus structure, phase recurrence, and numerical support

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1 Scope and status hierarchy

This supplement accompanies the shortened main-text version of the local Type-1, $N = 3$ selector reduction. It distinguishes clearly between theorem-level statements and numerically supported narrow assumptions.

Theorem-level results. At a simple transport ramification point of the canonical exact two-sheet reduction:

1. the exact local gauge is torus-valued to all orders;
2. the odd phase coefficients $D_{2m+1}(H)$ obey a compact Lagrange–Bürmann recurrence;
3. the full member-dependent phase series is affine-linear in $L_H(v)$ and $L'_H(v)$ across the commuting family;
4. the same-radius local exact-WKB selector multiplier is scalar.

Numerically supported narrow assumptions used only to descend back to the original outer frame.

Assumption 1 (outer torus survival)**.** *After spectator decoupling and subtraction of the universal curvature artifact, no nondecaying torus-breaking off-diagonal survives in the outer matcher.*

Assumption 2 (sector-independent residual constant part)**.** *The residual constant non-torus factor is sector-independent and geometry-only, so it may be absorbed into the resolved-sheet convention.*

The numerical basis for these assumptions is summarized in Sec. 7.

2 Exact two-sheet torus theorem

Let v be a simple transport ramification point, $u_* = u(v)$, and set $u = u_* + t^2$. The reduced sheet carriers satisfy exact scalar equations

$$\partial_u R_\pm = \mathrm{i}(E_\pm - E_x)R_\pm, \tag{1}$$

so on the t-cover,

$$\partial_t \begin{pmatrix} R_+ \\ R_- \end{pmatrix} = (A_v(t; H)I + B_v(t; H)\sigma_3) \begin{pmatrix} R_+ \\ R_- \end{pmatrix}, \tag{2}$$

with

$$A_v(t;H) = \mathrm{i}t\,(E_+ + E_- - 2E_x), \qquad B_v(t;H) = \mathrm{i}t\,(E_+ - E_-). \tag{3}$$

Thus the active local two-sheet system is exactly diagonal in the ordered resolved-sheet basis. Reconstruct the half-form amplitudes by

$$\Psi_\pm(t;H) = \sqrt{2t}\,\Gamma_\pm(t)R_\pm(t;H)\sqrt{\mathrm{d}t}. \tag{4}$$

After removing the common scalar phase, the exact local basis is

$$Y_v^{\mathrm{exact}}(t;H) = \mathrm{diag}\big(T_+(t)e^{\Phi_{\mathrm{rel}}(t;H)},\, T_-(t)e^{-\Phi_{\mathrm{rel}}(t;H)}\big), \tag{5}$$

where $T_\pm(t) = \sqrt{2t}\,\Gamma_\pm(t)$ and $\Phi_{\mathrm{rel}}(t;H) = \frac{\mathrm{i}}{2}S_{+-}(t;H)$. The WKB-normalized Airy basis preserves the same ordered line splitting $L_+ \oplus L_-$. Consequently the matching gauge is diagonal.

Theorem 1. *In the canonical resolved-sheet / WKB normalization, the exact local matching gauge has the form*

$$\widehat{G}_v(t;H) = \exp\!\big(\Theta_v(t;H)\sigma_3\big), \tag{6}$$

with analytic scalar exponent $\Theta_v(t;H)$. No torus-breaking off-diagonal term can survive in this canonical two-sheet reduction.

Proof. Both the exact and WKB-normalized local bases preserve the same ordered decomposition $L_+ \oplus L_-$. Any change of basis between two ordered bases of the same direct-sum splitting must preserve each line separately, hence is diagonal. Determinant-one normalization forces the form $\exp(\Theta_v\sigma_3)$. □

3 Lagrange–Bürmann recurrence for $D_{2m+1}(H)$

Write

$$z = \lambda - v, \qquad g_v(z) := u(v+z) - u_* = z^2 h_v(z), \qquad h_v(0) = \frac{u''(v)}{2} \neq 0. \tag{7}$$

Choose an analytic square root and define

$$\phi_v(z) := z\sqrt{h_v(z)}. \tag{8}$$

Since $\phi_v'(0) \neq 0$, it has an ordinary local inverse ψ_v, and the two inverse branches are

$$\lambda_+(t) = v + \psi_v(t), \qquad \lambda_-(t) = v + \psi_v(-t). \tag{9}$$

Set

$$F_H(z) := \Delta_H(v+z), \qquad \Delta_H(\lambda) = \frac{\mu(\lambda)L_H(\lambda)}{p(\lambda)}, \qquad \mu^2 = Q_4. \tag{10}$$

Define the odd coefficients by

$$\Delta_H(\lambda_+(t)) - \Delta_H(\lambda_-(t)) = 2\sum_{m\geq 0} D_{2m+1}(H)t^{2m+1}. \tag{11}$$

Ordinary Lagrange–Bürmann inversion yields

$$D_{2m+1}(H) = \frac{1}{2m+1}[z^{2m}]F_H'(z)\left(\frac{z}{\phi_v(z)}\right)^{2m+1}. \tag{12}$$

Because $\phi_v(z) = z\sqrt{h_v(z)}$, this becomes

$$D_{2m+1}(H) = \frac{1}{2m+1}[z^{2m}]F'_H(z)h_v(z)^{-(2m+1)/2}. \tag{13}$$

For computation, write

$$h_v(z) = h_0\left(1 + \sum_{k\geq 1} a_k z^k\right), \qquad h_0 = \frac{u''(v)}{2}, \tag{14}$$

$$Q_m(z) := \left(1 + \sum_{k\geq 1} a_k z^k\right)^{-(2m+1)/2} = \sum_{n\geq 0} q_n^{(m)} z^n. \tag{15}$$

Then coefficient comparison gives

$$q_0^{(m)} = 1, \tag{16}$$

$$q_n^{(m)} = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{1-2m}{2}k - n\right) a_k q_{n-k}^{(m)}. \tag{17}$$

If $F'_H(z) = \sum_{r\geq 0} \delta_r(H) z^r$, then

$$D_{2m+1}(H) = \frac{h_0^{-(2m+1)/2}}{2m+1}\sum_{r=0}^{2m} \delta_r(H) q_{2m-r}^{(m)}. \tag{18}$$

4 Affine-linearity on the commuting family

Because $\Delta_H = FL_H$ with $F(\lambda) = \mu(\lambda)/p(\lambda)$ and L_H linear, we have

$$L_H(v+z) = L_H(v) + L'_H(v)z. \tag{19}$$

Hence

$$F'_H(z) = L_H(v)F'_v(z) + L'_H(v)\big(F_v(z) + zF'_v(z)\big), \qquad F_v(z) = F(v+z). \tag{20}$$

Substituting into (13) or (18) yields

$$D_{2m+1}(H) = A_m(v)L_H(v) + B_m(v)L'_H(v), \tag{21}$$

with geometry-only sequences

$$A_m(v) = \frac{h_0^{-(2m+1)/2}}{2m+1}\sum_{r=0}^{2m}(r+1)F_{r+1}(v)q_{2m-r}^{(m)}, \tag{22}$$

$$B_m(v) = \frac{h_0^{-(2m+1)/2}}{2m+1}\sum_{r=0}^{2m}(r+1)F_r(v)q_{2m-r}^{(m)}. \tag{23}$$

Therefore the member-dependent phase series is an explicitly computable affine-linear sequence on the full commuting family.

5 Phase remainder and scalar multiplier

Since $\mathrm{d}u = 2t\,\mathrm{d}t$, the exact phase mismatch is

$$S_{+-}(t;H) = 4\sum_{m\geq 0} \frac{D_{2m+1}(H)}{2m+3} t^{2m+3}. \tag{24}$$

The linear Langer coordinate removes the cubic term exactly, so

$$\Theta_v^{\mathrm{ph}}(t;H) = 2\mathrm{i}\sum_{m\geq 1} \frac{D_{2m+1}(H)}{2m+3} t^{2m+3}. \tag{25}$$

If

$$\Theta_v^{\mathrm{ph}}(t;H) = \sum_{n\geq 2} \Xi_{2n+1}(H) t^{2n+1}, \tag{26}$$

then

$$\Xi_{2n+1}(H) = \frac{2\mathrm{i}}{2n+1} D_{2n-1}(H), \qquad n \geq 2. \tag{27}$$

Thus

$$\Xi_5(H) = \frac{2\mathrm{i}}{5} D_3(H), \qquad \Xi_7(H) = \frac{2\mathrm{i}}{7} D_5(H), \qquad \Xi_9(H) = \frac{2\mathrm{i}}{9} D_7(H), \ldots \tag{28}$$

Because the local inverse branches and compositions are analytic up to the nearest singularity, the local series is convergent there; the local Borel sum therefore coincides with the analytic function itself. In the WKB-normalized Airy convention, the active same-radius $S_1 \to S_0$ crossing carries the scalar multiplier

$$\mathfrak{s}_v^{\mathrm{BS}}(t;H) = -\mathrm{i}\exp\bigl(2\Theta_v(t;H)\bigr). \tag{29}$$

6 Canonical v_4 example and explicit t^7 result

For the standard real family

$$(\epsilon_0, \epsilon_1, \epsilon_2) = (-2, 0, 3), \qquad (\gamma_0, \gamma_1, \gamma_2) = \left(1, \frac{4}{5}, \frac{6}{5}\right), \tag{30}$$

at the canonical upper-half-plane selector point $v_4 = -0.8019106146 - 1.1076910061\,\mathrm{i}$, the recurrence reproduces the known Ξ_5 and yields explicit higher coefficients. In affine-linear form,

$$\Xi_7(H) = (-0.1218869377 + 0.5875264733\,\mathrm{i})\,\beta_H + (-0.5377433968 + 1.2623260628\,\mathrm{i})\,\alpha_H, \tag{31}$$

$$\Xi_9(H) = (-1.0157641727 + 0.8877006452\,\mathrm{i})\,\beta_H + (-2.5833648230 + 1.4527859580\,\mathrm{i})\,\alpha_H. \tag{32}$$

For the resonant wall member used in the local selector notes, this gives

$$\Xi_5(H_{\mathrm{wall}}) = 0.0511662458609726 + 0.106488885281404\,\mathrm{i}, \tag{33}$$

$$\Xi_7(H_{\mathrm{wall}}) = -0.0590453140023154 + 0.151640061843118\,\mathrm{i}, \tag{34}$$

$$\Xi_9(H_{\mathrm{wall}}) = -0.302262984767034 + 0.183667018646898\,\mathrm{i}. \tag{35}$$

Table 1: High-precision assays used to justify the outer canonical form.

Assay	Accepted samples	Outcome / interpretation
Resolved-sheet $\widehat{G}_v$ off-diagonal probe	1200	No off-diagonal counterexample found: 1200/1200 "no". Supports the exact two-sheet torus theorem from the numerical side.
Spectator-decoupled torus survival	1001 (301 targeted high-curvature)	1001/1001 pass at the main threshold. All 9/9 legacy high-curvature failures from the earlier raw assay pass on retest. Supports Assumption 1.
Sector-independence of residual constant part	1001 (301 targeted high-curvature)	999/1001 base-run passes; only 2 borderline extreme-curvature misses. Both pass refined 100-digit retest, leaving no durable counterexample. Supports Assumption 2.
Lagrange-inversion recurrence validation	1200	1200/1200 pass at 10^{-2} relative threshold and 1198/1200 at 10^{-3} for (D_1, D_3, D_5) simultaneously.

7 Numerical analyses that led to the outer canonical form

Table 1 records the numerical path that led to the outer canonical form. The theorem-level statements of the main text do not rely on these data; the data are used only to justify the narrow descent assumptions that identify the canonical two-sheet gauge with the original outer matcher.

Two details matter. First, the spectator-decoupled torus-survival assay intentionally oversampled the extreme-curvature regime, the most plausible source of counterexamples. Second, the two base-run sector-independence misses were not stable under refinement. Their refined retest values were:

sample	$\lvert u''(v)\rvert$	$\lvert c_0\rvert_{\text{refined}}$	fit error	pass at 10^{-6}?
810	370.801	7.14685e-07	3.53439e-07	yes
830	299.187	6.42155e-07	3.26301e-07	yes

These data strongly support the view that the surviving local structure in the original frame is torus-valued up to a sector-independent geometry-only constant factor.

8 What remains open and high-leverage tactics

The remaining unknowns are now sharply concentrated.

1. **Selector sufficiency.** The necessary sieve—transport ramification plus phase resonance—is formalized, but one still needs a proof that every simple phase-active transport ramification point actually inserts the scalar local block.
2. **Exact insertion rule.** The scalar local block is now explicit in the canonical two-sheet normalization. What remains is the exact local-to-global comparison theorem that inserts it into the continuation functor.

3. **Transport recurrence.** The phase side now has a closed affine-linear recurrence. A matching all-orders recurrence for the transport exponent $\Theta_v^{\mathrm{tr}}(t)$ would place the full torus exponent on the same algorithmic footing.
4. **Nongeneric collisions.** Degenerate transport ramification and common zeros of Q_4 and W_4 still require a separate analysis.

The highest-leverage next moves are symbolic: derive the transport recurrence, prove selector sufficiency in the scalarized two-sheet model, and then prove the exact global insertion rule. Numerics should now be used only as theorem-shaping tools: stress-test sufficiency hypotheses in the extreme-curvature regime, guess closed recurrences, and isolate the precise nongeneric hypotheses needed for theorem statements.

References

[1] *Multistate Landau–Zener Solver: Augmented ODE with Gauged Dynamics* (internal summary).

[2] *A note on the sextic gap polynomial for $N = 3$ Type-1 matrices* (internal note).

[3] *Type-1, $N = 3$ Landau–Zener primer: double cover, reduced ODE, gluing theorem, and selector* (internal primer).

[4] *Formalized strategy diagram for Type-1, $N = 3$ Landau–Zener* (internal note).

[5] *Local selector matching at a transport ramification point in Type-1, $N = 3$* (internal note).

[6] *Transport versus phase in the local gauge expansion at a simple transport ramification point* (internal note).